

FUTURE OF HIGHWAY CHARGES

- **After downloading the Global Navigation Satellite System (GNSS) tolling app, users will register their vehicles, link bank accounts and install an On-Board Unit (OBU), which transmits data to satellites**
- **The system tracks journeys, calculating entry/ exit points**
- **The GNSS system charges travellers for the exact distance travelled. Toll amount deducted from linked bank account**

in Karnataka and the Panipat-Hisar section of NH-709 in Haryana. The plan is to roll this out on 5,000 km of highways to begin with, including between Delhi and Mumbai. This week, 15 players, including International highway toll solutions companies, have submitted their expressions of interest to make the toll charger software. The ministry has formed an apex committee to oversee this.

However, implementing this is not without challenges. Mobile tower connectivity remains a concern in rural areas. Data security is another issue, as the system will handle vast amounts of user information. Additionally, there are fears about toll evasion and the technology's ability to accurately track and charge users.

The rollout will initially target commercial vehicles, with private cars to be included in phases. The system includes a centralised toll charger responsible for toll calculation, receiving data from on-board units (OBUs) fitted in vehicles. The government's concept paper says the GNSS system will integrate with existing FASTag infrastructure, using dedicated GNSS lanes at toll plazas initially, and will eventually transition all lanes to GNSS-based tolling. Major stakeholders include the National Highways Authority of India (NHAI), Indian Highway Management Company Ltd (IHMCL) and banks.

TELECOM CONNECTIVITY, DATA SECURITY AND TOLL EVASION ARE TESTS THE NEW SATELLITE SYSTEM FACE

But the rollout will take a while. "We are going by a timeline," says Vishal Chauhan, chairman, IHMCL, the agency implementing the project. "The request for proposal for appointing a toll charger entity will be issued this month. By December this entity will be in place.... By the end of this financial year, we will have the pilot in some of the stretches."

Toll collection is a subject fraught with controversies in India. High toll rates have often witnessed altercations, often violent, between motorists and toll plaza staff. Then there is the politics of it all. No one knows this better than Union transport minister Nitin Gadkari. "There are insiders in the system...they know how to switch off FASTags, shut down computers," Gadkari said at the stakeholder consultation in June. Ministry data shows India has around 127 million vehicles. But only 90 million pay toll, Gadkari said. "Around 25 per cent of the vehicles are not paying toll. They are stealing from the government." Currently, 97 per cent of toll amounts are being obtained through the tags. The government hopes that with GNSS the remaining three per cent can be mopped up too. The government earns around Rs 194 crore a day from 1,409 toll plazas across India.

But there are questions to be answered. For instance, what happens to all the vehicular data and protection of privacy? After the NHAI took a legal opinion from retired chief justice of India S.A. Bobde, it has been decided that the vehicle will be tracked only in the geo-zone of the toll plaza following user consent.

The most challenging aspect may well be outside the government's control: driving behaviour. The new system aims for Multi Lane Free Flow (MLFF), which means that vehicles are not supposed to stop while crossing a tolled road. But how will the system know from afar which vehicle does not have an OBU, or if the wallet has low balance? And how will non-GNSS vehicles be prevented from entering the dedicated lanes?

How Indian drivers will behave in the face of this new system initially is an area officials are deliberating. IIT-Delhi's Prof Geetam Tiwari for example has advised key performance indicators for device malfunctions, signage and overall management of lane changes at toll plazas to prevent high-speed accidents. In fact, it is being deliberated whether to have a barrier at the special GNSS lanes, or under which condition–like low balance or malfunctioning OBUs–the barrier will stop a vehicle.

"We cannot blindly copy from the West because, frankly, their full-scale deployment is equivalent to our pilot. We need to develop our own systems based on our own unique requirements," says Akhilesh Srivastava, International Transport Forum, India.

That is, both operational and theoretical doubts remain, but the government has set off the process where long queues at toll plazas–indeed, toll plazas themselves–will be a thing of the past. ■